

CPR 2 MODULE OBJECTIVES				
Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 25	HCB	Histology of Lymphatic Tissues & Organs	SOM.MK.1.BPM1.3.CPR.1.HCB.1224	Define lymphatic tissue using the concepts of BALT, MALT and GALT.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1225	Differentiate between primary and secondary lymphatic nodules based on their histological structure.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1226	Compare primary vs. secondary lymphoid organs.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1227	Give the function(s) of the thymus.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1228	Describe the histological structure of the thymus.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1229	State the significance of the blood thymus barrier.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1230	Describe the histological structure of the blood thymus barrier.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1231	State the function of a lymph node.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1232	Describe the histological structure of a lymph node including the cellular distribution.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1233	Describe the flow of lymph through a lymph node.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1234	Explain the significance of the subcapsular sinus in cancer metastasis through a lymph node.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1235	Explain the significance of the stroma in a lymph node.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1236	State the function(s) of the spleen.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1237	Describe the histological structures of the spleen including the organization of the white and red pulp.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1238	Describe the general circulation through the spleen.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1239	Microscopically identify the white pulp.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1240	Give the function of the white pulp
			SOM.MK.1.BPM1.3.CPR.1.HCB.1241	Microscopically identify the red pulp.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1242	Give the function of the red pulp
			SOM.MK.1.BPM1.3.CPR.1.HCB.1243	Describe the importance of the marginal zone.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1244	Describe the importance of the periarteriolar lymphatic sheath (PALS).
			SOM.MK.1.BPM1.3.CPR.1.HCB.1245	Distinguish between the three (3) types of tonsils based on their location and microscopic structure.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1246	Describe the concept of Waldeyer's lymphatic ring.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1247	Define the term "lymphadenopathy".
			SOM.MK.1.BPM1.3.CPR.1.HCB.1248	Explain the consequence of thymic aplasia in Di George syndrome.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1249	Correlate the histologic changes that take place in the spleen secondary to its enlargement in pathological states.
Lecture 26	ANAT	Development and Clinical Embryology of the Respiratory System	SOM.MK.II.BPM1.3.CPR.2.ANAT.1428	List the germ layers contributing to the formation of the respiratory tract.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1429	Describe the development of the larynx, trachea, bronchial tree and lungs.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1430	Describe the stages of lung development.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1431	Name the site of production and the role of surfactant.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1432	Summarize the changes that occur in the respiratory system at birth.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1433	Describe the development of the diaphragm.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1434	Describe the division of the embryonic coelom into pleural, pericardial and peritoneal cavities.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1435	Explain the embryological mechanism for all types of congenital diaphragmatic herniae, and eventration.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1436	Explain the embryological mechanism for all types of tracheoesophageal fistulas.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1437	Describe how tracheoesophageal fistula can lead to polyhydramnios.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1438	Explain the embryological basis for laryngeal atresia, tracheal stenosis and atresia, and esophageal atresia and stenosis.
Lecture 27	ANAT	Clinical Anatomy of the Respiratory System	SOM.MK.II.BPM1.3.CPR.2.ANAT.1439	Describe the locations of the parietal and visceral pleura and the areas where they meet.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1440	List the recesses formed by the pleural reflections.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1441	Differentiate between the thoracic cavity and the pleural cavity.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1442	Describe the relationship of the lungs to the pleural cavity.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1443	Describe and differentiate the surface projections (rib level) of the visceral pleura and those of the parietal pleura.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1444	List the identifying features of the right and left lungs and their surface projections.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1445	List the surfaces and borders of the lungs.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1446	Describe the surface projections (rib level) of lobes and fissures of the lungs.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1447	Identify the auscultation sites for lobes of the lung.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1448	Differentiate between the root and hilum of a lung.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1449	Describe the relationship of the structures in the root of each lung.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1450	Describe the typical branching pattern of the bronchial tree.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1451	Describe the arterial supply, venous and lymphatic drainage of the lungs and the bronchial tree.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1452	Describe a bronchopulmonary segment and its clinical importance
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1453	Describe the autonomic innervation of the lungs and bronchial tree including the specific response of each structure (glands, bronchial smooth muscle, and vascular smooth muscle) to the sympathetic and parasympathetic innervation respectively.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1454	Identify the major anatomical features of the lungs in imaging studies including but not limited to CT, MRI and radiographic images.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1455	Describe the embryological basis and clinical presentation of respiratory distress syndrome (hyaline membrane disease).

			SOM.MK.II.BPM1.3.CPR.2.ANAT.1456	Compare and contrast the mechanism underlying a spontaneous vs traumatic pneumothorax.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1457	Compare and contrast the mechanism underlying an open pneumothorax vs a tension pneumothorax.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1458	Describe the clinical signs of a tension pneumothorax and explain why it is a medical emergency.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1459	Describe the decompression technique for a tension pneumothorax.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1497	Differentiate the radiological features of pneumothorax, hemothorax (pleural effusion), and hemo-pneumothorax (hydro-pneumothorax).
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1460	Describe the location of pleural effusion versus pulmonary edema.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1461	Explain why an aspirated foreign body would usually pass into the right bronchus.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1462	Describe atelectasis.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1463	Describe the anatomical location, mechanism and clinical sequences of the insertion of an endotracheal tube.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1496	Describe a Pancoast tumor and explain how it can lead to thoracic outlet syndrome.
DLA 13	HCB	Respiratory System	SOM.MK.I.BPM1.3.CPR.1.HCB.1254	List the divisions of the respiratory system.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1255	Describe the components of each division of the respiratory system.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1256	Describe the 5 cell types of the respiratory epithelium.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1257	Describe the histological structure of the vestibular and vocal folds of the larynx
			SOM.MK.I.BPM1.3.CPR.1.HCB.1258	State the function(s) of the larynx
			SOM.MK.I.BPM1.3.CPR.1.HCB.1259	Predict the outcome of defective cilia in Kartagener's Syndrome in the respiratory system.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1260	Describe the histopathogenesis of cystic fibrosis specific to the respiratory system.
Lecture 28	HCB	Histology of the Respiratory System	SOM.MK.I.BPM1.3.CPR.1.HCB.1261	Differentiate between the three regions of the nasal cavity based on the mucosa and structure.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1262	Describe the layers, structures and functions of the trachea. Note that the primary bronchi carry the same histological features as the trachea.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1263	Describe the microscopic structure of a secondary bronchi.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1264	Describe the microscopic structure of a tertiary bronchi.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1265	Give the function(s) of bronchiole.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1266	Describe the microscopic appearance of the structure of bronchioles.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1267	Differentiate between terminal and respiratory bronchioles based on location and microscopic structure.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1268	Describe the function(s) of club (Clara) cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1269	Discuss the significance of club (Clara) cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1270	Identify the function of alveolar ducts and alveolar sacs.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1271	Describe the microscopic structure of alveolar ducts.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1272	Describe the microscopic structure of alveolar sacs.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1273	State the function(s) of alveoli.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1274	Describe the histological appearance of the alveoli.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1275	State the function(s) of surfactant.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1276	Describe the microscopic structure of the interalveolar septum.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1277	State the significance of the interalveolar septum.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1278	State the function of the blood-air barrier.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1279	Describe the microscopic structure of the blood-air barrier.
Lecture 29	HCB	Flipped Class: Inflammation	SOM.MK.I.BPM1.3.CPR.1.HCB.1280	State the clinical significance of pores of Kohn in the spread of infections within the lungs.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1281	Identify the regions of the respiratory tract affected in asthma, chronic bronchitis, emphysema and pneumonia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1287	State the importance of inflammation
			SOM.MK.I.BPM1.3.CPR.1.HCB.1288	List the two major categories of inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1289	Describe the three major components of the acute inflammatory exudate.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1290	Discuss the mode of formation of an acute inflammatory exudate.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1291	List the vasoactive mediators involved in various stages of the acute inflammatory response.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1292	List the chemical mediators involved in various stages of acute inflammatory response.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1293	Describe the three morphological types of acute inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1294	Give an example of each of the morphological types of acute inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1295	Define the terms "pus", "abscess" and "pyogenic bacteria".
			SOM.MK.I.BPM1.3.CPR.1.HCB.1296	Describe the three major components of the acute inflammatory transudate.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1297	Discuss the mode of formation of the acute inflammatory transudate.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1298	List and describe the cardinal signs of inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1299	List and describe the three possible outcomes of acute inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1300	Describe the cell types that are predominant in acute inflammatory infiltrates.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1301	Describe the main factors that determine the outcome of an acute inflammatory condition.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1302	Describe the role of granulation tissue in wound healing and resolution of acute inflammation.

			SOM.MK.I.BPM1.3.CPR.1.HCB.1303	Describe the cell types that are predominant in chronic inflammatory infiltrates.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1304	Describe the typical outcome of chronic inflammation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1305	Define granuloma.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1306	Describe 2 different types of granuloma formation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1307	Identify the predominant cell type in granulomas.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1308	Compare and contrast granulation tissue vs. granuloma.
DLA 14	PHYS	Lung Function and Respiratory Gases	SOM.MK.II.BPM1.3.CPR.2.PHYS.0191	Describe the structural characteristics of airways.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0192	Describe the transport of gas through the conducting airways to and from the alveoli.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0193	Explain the key roles of mucociliary transport and describe related pathologies and their clinical presentation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0194	Describe the clinical significance of mucociliary transport.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0195	Describe the functions of alveolar Type I & Type II cells.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0196	List components of chest wall and lungs and relate the functions of respiratory muscles to movement of air in and out of the alveoli.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0197	Describe how Boyle's and Charle's law apply to movement of gas in and out of lungs.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0198	List the metabolic functions of the lungs and explain their key physiological roles.
Lecture 30	PHYS	Respiratory Muscles and Breathing Cycle	SOM.MK.II.BPM1.3.CPR.2.PHYS.0199	Describe how pleural pressure, alveolar pressure, airflow, lung recoil and lung volume change during a normal quiet breathing cycle or a deep breathing cycle with forced expiration.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0200	Identify the points of onset of inspiration, cessation of inspiration, and cessation of expiration in a figure representing Intrapleural pressure, airflow and alveolar pressure changes.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0201	Describe how differences in pressure between the atmosphere and alveoli cause air to move in and out of the lungs.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0202	Describe pulmonary pressure-volume (compliance) curve, inflation and deflation limbs. Explain the cause and significance of the hysteresis in the curves.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0203	Describe the factors that alter lung compliance in most common lung diseases.
DLA 15	PHYS	Airflow and Airways	SOM.MK.II.BPM1.3.CPR.2.PHYS.0208	Describe the effects of airway diameter and turbulent flow on airway resistance.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0209	Explain dynamic compression of the airways.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0210	Use the concept of dynamic compression of airways to explain why each point in the effort- independent region of the curve represents a maximal flow rate that is uniquely dependent on lung volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0211	Describe how and why the shape of the flow-volume curve is shifted in chronic obstructive or restrictive lung disease (COPD).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0212	Explain how mechanical changes with restrictive and obstructive disease affect the EPP.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0364	Explain the equal pressure point (EPP) and the forces that contribute to its formation during a forced expiration.
DLA 16	PHYS	Pulmonary Circulation	SOM.MK.II.BPM1.3.CPR.2.PHYS.0213	Contrast the systemic and pulmonary circulations with respect to pressures, resistance to blood flow, and response to hypoxia.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0214	Explain the regional differences in pulmonary blood flow in an upright person. Define zones I, II, and III in the lung, with respect to pulmonary vascular pressure and alveolar pressure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0215	Explain how distention and recruitment of pulmonary vessels influences pulmonary vascular resistance with alterations in cardiac output or pulmonary arterial pressure. Explain in terms of distention and recruitment of pulmonary vessels. Identify the zones in which these two mechanisms apply.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0216	Explain how pulmonary vascular resistance in alveolar and extra-alveolar blood vessels changes with lung volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0217	Describe the consequence of hypoxic pulmonary vasoconstriction on the distribution of pulmonary blood flow.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0218	Explain the development of pulmonary edema by a) increased hydrostatic pressure, b) increased permeability, c) impaired lymphatic outflow or increased central venous pressure, and d) hemodilution (e.g., with saline volume resuscitation).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0219	Define right-to-left shunts, anatomic and physiological shunts, and physiologic dead space.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0365	Describe the consequences of each of the above in pulmonary gas exchange.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0221	Draw a normal spirogram, labeling the four lung volumes and four capacities. List the volumes that comprise each of the four capacities. Identify which volume and capacities cannot be measured by spirometry.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0222	Indicate the value of using the Helium Dilution test and Plethysmography.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0223	Explain the factors that determine total lung capacity, functional residual capacity, and residual volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0366	Describe the mechanisms responsible for the changes in the lung volumes and capacities that occur in patients with emphysema and pulmonary fibrosis.

Lecture 31	PHYS	Spirometry	SOM.MK.II.BPM1.3.CPR.2.PHYS.0224	Draw a spirogram resulting from a maximal expiratory effort and label the forced vital capacity (FVC), timed forced expiratory volumes (FEVs), and the maximal expiratory flow rate between 25-75% of FVC (FEF _{25-75%}).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0367	Describe how the mechanical forces that contribute to the pulmonary function indices alter with obstructive or restrictive disease.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0225	Identify the effort-dependent and -independent regions on a normal maximal effort flow-volume curve.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0368	Describe how dynamic compression of airways influences the shape of the effort-independent region of the maximal effort flow-volume curve.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0369	Describe how and why the shape of the flow-volume curve is shifted in chronic obstructive and restrictive lung diseases.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0226	Differentiate between the two broad categories of restrictive and obstructive lung disease, including the spirometric abnormalities associated with each category.
Lecture 32	BCHM	Hemoglobin and Gas Transport	SOM.MK.I.BPM1.1.CPR.2.BCHM.0286	Analyze how hemoglobin can regulate its oxygen affinity to pick up oxygen in the lungs, and deliver oxygen to the tissues
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0287	Discuss the structure-function relationships in disorders associated with hemoglobin.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0288	Discuss fetal hemoglobin and explain why it has a higher affinity for oxygen compared to HbA
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0289	Distinguish between hemoglobin (Hb) and myoglobin structures.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0290	Describe the conformational change in the Hb molecule following oxygen binding.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0291	Outline how hemoglobin participates in oxygen and carbon dioxide transport in blood
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0292	Discuss how carbon monoxide (CO) binds to Hb and how this interferes with normal oxygen binding.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0293	Discuss the significance of the sigmoidal oxygen binding curve of Hb, and how this is produced by cooperative binding of O ₂ to Hb.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0294	Describe the function of 2,3-BPG in regulating oxygen binding to Hb. Analyze the role of 2,3 BPG in adaptation to high altitude, and its role in oxygen binding to fetal hemoglobin, and in blood transfusion.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0295	Explain the Bohr effect and its significance in regulating oxygen release from oxyHb (H ⁺ , CO ₂).
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0296	Compare and contrast the structure of HbA, HbA ₂ , HbA ₁ C and HbF.
			SOM.MK.I.BPM1.1.CPR.2.BCHM.0297	Describe methemoglobin and its impact on the structure and function of hemoglobin
Lecture 33	GNET	Hemoglobinopathies	SOM.MK.III.BPM1.1.CPR.3.GNET.0180	Review molecular mechanism of sickle cell anemia from the FTM module.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0181	Interpret data and differentiate between RFLP and ASO blot in the interpretation of inherited hemoglobin disorders.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0182	Explain the effect on the protein due to a single base substitution in HbS and HbC.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0183	Analyze the difference between sickle cell anemia (homozygous) and sickle cell trait (heterozygous/carrier). Indicate the role of heterozygote advantage of the sickle cell trait in parts of the world where malaria is endemic
			SOM.MK.III.BPM1.1.CPR.3.GNET.0184	Explain the molecular basis for the clinical features of HbS and HbC. Discuss the basis of hydroxyurea in management of HbS
			SOM.MK.III.BPM1.1.CPR.3.GNET.0185	Discuss the HbC mutation and the phenotype of the HbC carrier, HbC homogygotes and HbS/HbC individuals.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0186	Compare the genomic arrangement of the α -globin and β -globin genes
			SOM.MK.III.BPM1.1.CPR.3.GNET.0187	Compare, and contrast the most common mutations in α -thalassemia and β -thalassemia, explaining the thalassemia trait (heterozygous state).
			SOM.MK.III.BPM1.1.CPR.3.GNET.0188	Explain the concept of allelic heterogeneity and compound heterozygote in relation to β -thalassemia.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0189	Analyze the genotype-phenotype correlations in α -thalassemia and β -thalassemia.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0190	Explain the mode of inheritance of hemophilia A and B, the common clinical features and the most common mutation (Intron inversion) in hemophilia A.
			SOM.MK.III.BPM1.1.CPR.3.GNET.0191	Explain the genetic basis of manifestations in a hemophilia carrier female (skewed X inactivation)
			SOM.MK.III.BPM1.1.CPR.3.GNET.0192	Calculate the recurrence risk in families with hemophilia A (When mother is a carrier or when father is affected), HbS, HbC, alpha-thalassemia and beta-thalassemia
Lecture 34	PHYS	Mechanics of Lung and Chest	SOM.MK.II.BPM1.3.CPR.2.PHYS.0204	Explain surface tension and how it applies to lung mechanics, including the effects of alveolar size and the role of surfactants. Infer the role of surfactants in preventing atelectasis.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0205	Illustrate and explain the pressure-volume (compliance) curves for the lungs, chest wall, and respiratory system on the same set of axes. Show and explain the significance of the resting positions for each of these three structures.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0206	Identify the forces that generate the negative intrapleural pressure when the lung is at functional residual capacity, and interpret the direction that the lung and chest wall will move if air is introduced into the pleural cavity (pneumothorax).

			SOM.MK.II.BPM1.3.CPR.2.PHYS.0207	Explain how lung and chest wall compliance interact to set FRC and how changes in lung or chest wall recoil can change in a patient with obstructive/restrictive disease or with pneumothorax.
DLA 17	PHYS	Regulation of Breathing	SOM.MK.II.BPM1.3.CPR.2.PHYS.0227	Identify the regions in the central nervous system that play important roles in the generation and control of cyclic breathing.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0228	Give three examples of reflexes involving pulmonary receptors that influence breathing frequency and tidal volume. Describe the receptors and neural pathways involved.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0229	List the anatomical locations of chemoreceptors sensitive to changes in arterial PO ₂ , PCO ₂ , and pH that participate in the control of ventilation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0372	Differentiate the relative importance of each in sensing alterations in blood gases.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0230	Describe how changes in arterial PO ₂ and PCO ₂ alter alveolar ventilation, including the synergistic effects when PO ₂ and PCO ₂ both change.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0231	Explain the respiratory drive in a COPD patient.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0373	Explain the change in respiratory drive when oxygen is given to a COPD patient.
Lecture 35	PHYS	Ventilation	SOM.MK.II.BPM1.3.CPR.2.PHYS.0232	Define partial pressure and fractional concentration as they apply to gases in air. List the normal fractional concentrations and sea level partial pressures for O ₂ , CO ₂ , and N ₂ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0233	List the normal airway, alveolar, arterial, and mixed venous PO ₂ and PCO ₂ , O ₂ saturation, [HCO ₃ ⁻] and pH.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0234	Define and contrast the following terms: anatomic dead space, physiologic dead space, wasted (dead space) ventilation, total minute ventilation and alveolar minute ventilation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0235	Describe the concept by which physiological dead space can be measured.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0236	Explain the relationship between alveolar ventilation and the arterial PCO ₂ and PO ₂ and pH, compliance and ventilation. Describe how hypoventilation would affect alveolar and arterial PO ₂ and PCO ₂ and the A-a gradient.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0237	Describe in quantitative terms the effect of ventilation on PCO ₂ according to the alveolar ventilation equation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0238	Calculate the alveolar oxygen partial pressure (PAO ₂) and the the amount of supplemental O ₂ required to overcome a reduction in PAO ₂ caused by hypoventilation or high altitude using the simplified form of the alveolar gas equation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0239	Define the following terms: hypoventilation, hyperventilation, hypercapnea, eupnea, hypopnea, and hyperpnea.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0240	Calculate the alveolar to arterial PO ₂ difference (A-a) DO ₂ . List the normal value for (A-a) DO ₂ and explain the clinical significance of an elevated (A-a) DO ₂ .
Lecture 36	BCHM	Flipped Class: Glycolysis & Hemolysis	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0298	Define glycolysis and explain its role in the generation of metabolic energy.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0299	List the reactions and enzymes that convert glucose into pyruvate (emphasize key enzymes and irreversible reactions)
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0300	Define substrate level phosphorylation and identify those reactions in glycolytic pathway
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0301	Explain significance of lactate production in anaerobic glycolysis and describe the fate of lactate formed in RBC and muscle (Cori cycle)
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0302	Outline the regulation of glycolysis indicating the regulatory enzymes and highlight the role of AMP, ATP on glycolysis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0303	Review the significance of glucose transporters (GLUT1-5) present in various tissues
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0304	Compare and contrast the action of glucokinase and hexokinase (with the help of a graph)
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0305	Review the regulation of glycolysis
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0306	Outline the function of glycolysis in specific tissues (RBC) including 2,3-BPG formation and energy
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0307	Analyze the effect of fluoride ions on glycolytic enzymes
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0308	Identify the biochemical mechanisms involved in the occurrence of lactic acidosis in states of poor tissue perfusion/ oxygenation.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0309	Explain the basis for occurrence of hemolysis in inherited pyruvate kinase deficiency
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0310	Compare the changes in 2,3 BPG levels and hemoglobin oxygen dissociation curve in hexokinase and pyruvate kinase deficiency
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0311	Differentiate between pyruvate kinase deficiency (defect in glycolytic pathway) and glucose 6-phosphate dehydrogenase deficiency (pentose phosphate defect)
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0246	Define the ventilation/perfusion (V/Q) ratio of an alveolar-capillary lung unit and show how it determines the PO ₂ and PCO ₂ of the blood emerging from that lung unit.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0247	Calculate the average V/Q ratio in a normal lung and explain how V/Q is affected by the vertical distribution of ventilation and perfusion in the healthy lung.

Lecture 37	PHYS	Ventilation Perfusion Inequality and Hypoxemia	SOM.MK.II.BPM1.3.CPR.2.PHYS.0248	Predict how the presence of abnormally low and high V/Q ratios in a person's lungs will affect arterial PO ₂ and PCO ₂ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0249	Describe two causes of abnormal V/Q distribution.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0250	Explain the right-to-left shunts, anatomic and physiological shunts, and physiologic dead space (wasted ventilation) and the consequences of each for pulmonary gas exchange.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0251	Explain the airway and vascular control mechanisms that help maintain a normal ventilation/perfusion ratio. Name two compensatory reflexes for V/Q inequality.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0252	Calculate the alveolar to arterial PO ₂ difference, (A-a) O ₂ gradient. List the normal value range for (A-a) O ₂ and explain the significance of an elevated (A-a) O ₂ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0253	Describe how changes to the A-a gradient occur and what factors lead to an increased A-a gradient in patients.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0254	Describe the mechanisms for the shift in alveolar ventilation that occur immediately upon ascent to high altitude, after remaining at altitude for two weeks, and immediately upon return to sea level.
Lecture 38	PHYS	Gas Transport	SOM.MK.II.BPM1.3.CPR.2.PHYS.0255	Define oxygen partial pressure (tension), oxygen content, and percent hemoglobin saturation as they pertain to blood.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0256	Draw an oxyhemoglobin dissociation curve (hemoglobin oxygen equilibrium curve) showing the relationships between oxygen partial pressure, hemoglobin saturation, and blood oxygen content.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0257	On the dissociation curve, illustrate the relationship between PO ₂ and dissolved plasma O ₂ content. Contrast the relative amounts of O ₂ carried bound to hemoglobin with that carried in dissolved form.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0258	Describe how the shape of the oxyhemoglobin dissociation curve influences the uptake and delivery of oxygen and SaO ₂ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0259	Define P ₅₀ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0260	Show how the oxyhemoglobin dissociation curve is affected by changes in blood temperature, pH, PCO ₂ , and 2,3-DPG, and describe a situation where such changes have important physiological consequences.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0261	Describe how anemia and carbon monoxide poisoning affect the shape of the oxyhemoglobin dissociation curve, PaO ₂ .
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0262	Name the factors that affect diffusive transport of a gas between alveolar gas and pulmonary capillary blood.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0263	Describe the kinetics of oxygen transfer from alveolus to capillary and the concept of capillary reserve time (i.e., the portion of the erythrocyte transit time in which no further diffusion of oxygen occurs).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0264	Define oxygen diffusing capacity, and describe the rationale and technique for the use of carbon monoxide to determine diffusing capacity.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0265	Calculate the diffusing capacity of carbon monoxide (DLCO) from the alveolar to arterial PO ₂ difference. Describe the normal value for DLCO and the significance of a decreased DLCO.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0266	List the forms in which carbon dioxide is carried in the blood. Identify the percentage of total CO ₂ transported as each form.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0267	Describe the importance of the chloride shift in the transport of CO ₂ by the blood.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0268	Identify the enzyme that is essential to normal carbon dioxide transport by the blood and its location.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0269	Explain the carbon dioxide dissociation curves for oxy- and deoxyhemoglobin. Describe the interplay between CO ₂ and O ₂ binding on hemoglobin that causes the Haldane effect.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0270	Define respiratory acidosis and alkalosis and give clinical examples of each.
Lecture 39	PHYS	Flipped Class: Respiratory Case Analyses	SOM.MK.II.BPM1.3.CPR.2.PHYS.0337	Discuss pathophysiologic basis of clinical features, laboratory findings and complications in a patient with chronic obstructive lung disease
Lecture 40	ANAT	Development and Clinical Embryology of the Urinary System	SOM.MK.II.BPM1.3.CPR.2.ANAT.1464	Describe the development, location, blood supply, innervation and venous drainage of the suprarenal glands.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1465	Describe the development of the kidney from the metanephric diverticulum and the metanephric mass of mesoderm.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1466	Describe the development of the ureters.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1467	Describe the development of the urinary bladder and urethra.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1468	Describe the development of a discoid (pancake) and a horseshoe kidney.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1469	Describe the development of a bifid or double ureter.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1470	Describe the embryologic mechanism responsible for renal agenesis.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1471	Describe how a kidney may have more than one renal artery and how an accessory renal artery may cause hydronephrosis.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1358	Describe the location of the kidneys and their relationship with surrounding organs and ribs 11 and 12.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1359	Describe the relationship of the kidneys to the diaphragm, psoas major, quadratus lumborum and transversus abdominis aponeurosis.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1360	Describe the anatomy of the cortex and medulla of the kidney.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1361	Identify the structures in the hilum of the kidney and their relationship to one another.

DLA 18	ANAT	Clinical Anatomy of the Urinary System - Upper Urinary Tract	SOM.MK.II.BPM1.3.CPR.2.ANAT.1362	Describe the blood supply, innervation and venous drainage of the kidney.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1472	Identify the course, relationships, blood supply and innervation of the ureters.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1473	Describe the kidneys in relation to the renal fascia, perirenal and pararenal fat.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1475	Describe the course of the ureter along the posterior abdominal wall to the bladder.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1477	Describe the structures which support the urinary bladder.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1478	Describe the internal anatomy of the urinary bladder.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1479	Describe the blood supply, venous drainage, lymphatic drainage and innervation of the urinary bladder.
Lecture 41	ANAT	Flipped Class: Clinical Anatomy of the Urinary System - Lower Urinary Tract and Clinical Correlates	SOM.MK.II.BPM1.3.CPR.2.ANAT.1474	Describe “the water under the bridge” relationship
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1476	List the sites of constriction of the ureters and explain their relationship to kidney stones
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1480	Describe the flow of urine from the renal papillae to the ureter and to the urinary bladder.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1481	Describe the retropubic space.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1482	Compare and contrast the male versus female urethra.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1483	Compare the innervation of the internal and external urethral sphincters.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1484	Describe the muscles and nerves involved in normal micturition and urinary continence.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1485	Explain why the empty bladder is a pelvic organ in adolescents and adults, but a partially abdominal organ in infants and small children (SG) .
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1486	Describe suprapubic catheterization and explain how a needle inserted through the lower abdominal wall over the pubic symphysis can enter a distended bladder without entering the peritoneal cavity (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1487	Identify, in radiological images, the major organs of the renal system including the kidneys, ureters and bladder
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1488	Define the term “cystocele” and explain why it is more common in multiparous women.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1489	Define the term “stress urinary incontinence” and explain which structures may be injured during vaginal delivery to lead to this condition.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1491	Explain the distribution of pain from kidney stones that are impacted (lodged) in the ureter.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1492	Describe the layers that would be cut when harvesting a donor kidney using a posterior approach.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1493	Describe the anatomical location, mechanism and clinical consequences of renal vein entrapment syndrome.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1494	Describe the development of cystic kidney disease.
Lecture 42	HCB	Histology of the the Nephron & Blood Pressure Control Mechanisms	SOM.MK.1.BPM1.3.CPR.1.HCB.1309	List at least 4 major functions of the kidneys.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1310	Briefly recap the gross anatomical structures of a kidney.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1311	Differentiate between the renal columns (of Bertin), renal pyramids, cortical labyrinth, renal lobes and lobules based on their components and location.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1312	Describe the components of the renal cortex.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1313	Describe the components of the renal medulla.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1314	Describe the flow of blood through the kidney.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1315	Distinguish between a cortical, intermediate and juxtamedullary nephron based on their structure and location.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1316	Identify the components of a urinereforus tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1317	List the components of a renal nephron (including the renal corpuscle and tubules).
			SOM.MK.1.BPM1.3.CPR.1.HCB.1318	State the function of the renal corpuscle.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1319	Describe the microscopic structure of the renal corpuscle.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1320	Discuss the structural significance of glomerular capillaries.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1321	Discuss the significance of the podocytes.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1322	Identify the location of intra-glomerular mesangial cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1323	Explain the function(s) of intraglomerular mesangial cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1324	Describe the structures of the glomerular filtration barrier.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1325	Explain the significance of the glomerular filtration barrier
			SOM.MK.1.BPM1.3.CPR.1.HCB.1326	Describe the flow of urinary filtrate in sequence from Bowman’s space to the exterior.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1327	Identify the function(s) of the proximal convoluted tubules.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1328	Describe the microscopic structure of the proximal convoluted tubules.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1329	Describe the histological changes in the glomerular basement membrane in antibody-induced glomerulonephritis, Goodpasture syndrome and Alport’s syndrome.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1330	State the functions of the proximal straight tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1331	Describe the microscopic structure of the proximal straight tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1332	Give the function(s) of the thin segment of descending and ascending loops of Henle.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1333	Describe the microscopic structure of the thin segment of descending and ascending loops of Henle.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1334	Give the function of the vasa rectae.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1335	Explain the functional significance of the vasa rectae.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1336	Give the function of the distal straight tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1337	Describe the microscopic structure of the distal straight tubule.

Lecture 43	HCB	Histology of the Excretory Part of the Renal System	SOM.MK.1.BPM1.3.CPR.1.HCB.1338	Give the function of the distal convoluted tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1339	Describe the microscopic structure of the distal convoluted tubule.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1340	Explain the clinical significance of each component of the juxtaglomerular apparatus.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1341	Identify the location of the juxtaglomerular apparatus.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1342	Describe the site of production of renin.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1343	Discuss the role of renin in the renin-angiotensin-aldosterone system.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1344	Give the function of connecting tubules.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1345	Describe the microscopic structure of connecting tubules.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1346	Distinguish between the cortical and medullary collecting ducts based on their microscopic structure, function and location.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1347	Describe the microscopic structure of the papillary ducts (of Bellini).
			SOM.MK.1.BPM1.3.CPR.1.HCB.1348	State the functional significance of the papillary ducts
			SOM.MK.1.BPM1.3.CPR.1.HCB.1349	Describe the contents of the renal interstitium.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1350	Describe the microscopic structure of the minor calyx.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1351	Describe the microscopic structure of the major calyx.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1352	Explain the clinical significance of the junctional complexes of the urothelium.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1353	Give the function of the ureter.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1354	Describe the microscopic structure of the ureter.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1355	Give the function of the urinary bladder.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1356	Describe the microscopic structure of the urinary bladder.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1357	Compare the male and female urethra based on length, location and epithelial lining.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1358	Identify the renal segments that respond to the following hormones: atrial natriuretic factor, parathyroid hormone, anti diuretic hormone and aldosterone, and the effects of each hormone on the segments.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1359	Describe the role of the renin-angiotensin-aldosterone system (RAAS) in Na ⁺ homeostasis.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1360	Describe the role of the renin-angiotensin-aldosterone system (RAAS) in blood volume homeostasis.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1361	Describe the role of the renin-angiotensin-aldosterone system (RAAS) in blood pressure regulation.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1362	State the role of antidiuretic hormone (ADH) in the regulation of the function of the collecting duct.
Lecture 44	PHYS	Glomerular Mechanisms	SOM.MK.II.BPM1.3.CPR.2.PHYS.0276	Identify the filtration barriers, if any, which impede the filtration of H ₂ O, Na ⁺ , inulin, albumin, and red blood cells.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0277	Define the filtration coefficient at the glomerular capillary, describe the membrane properties that contribute to it, and explain its role in determining GFR.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0278	Given the capillary and Bowman's capsule hydrostatic and oncotic pressures, calculate the net filtration force at the glomerular capillaries.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0352	Predict the changes in glomerular filtration caused by increases or decreases in any of those pressures.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0284	Define renal blood flow, renal plasma flow, glomerular filtration rate, and filtration fraction and list typical values.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0279	Describe the relative resistances of the afferent and efferent arterioles and the effects on renal blood flow and GFR of selective changes in each.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0280	Describe the myogenic and tubuloglomerular feedback mechanisms that mediate the autoregulation of renal plasma flow and glomerular filtration rate.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0281	Predict the changes in net filtration force that occur as blood travels along the glomerular capillary and colloid osmotic pressure increases.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0282	Explain the effects of changes in peritubular capillary hydrostatic and colloid osmotic pressures on net proximal tubular fluid reabsorption.
Lecture 45	PHYS	Tubular Mechanisms	SOM.MK.II.BPM1.3.CPR.2.PHYS.0283	Define and understand the concepts of net transport: filtration, reabsorption, secretion, excretion.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0285	Distinguish between active (primary and secondary) transport, facilitated diffusion, and passive diffusion based on energy source and carrier protein involvement.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0286	Explain the contribution of the major nephron segments (PT, LOH, DCT, CT) to the reabsorption of the filtered load of solutes and water.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0287	Describe the cellular mechanisms for the transport of Na ⁺ , Cl ⁻ , K ⁺ , HCO ₃ ⁻ , Ca ²⁺ , phosphate, organic solutes (e.g., glucose, amino acids, and urea), and water by the major tubular segments.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0288	Describe the function of the following renal transporters and their predominant localization along the tubules with regard to nephron segment and apical versus basolateral membranes: a. Transport ATPases (Na ⁺ /K ⁺ -ATPase, H ⁺ /K ⁺ -ATPase, H ⁺ -ATPase, and Ca ²⁺ -ATPase); b. Ion and water channels (K ⁺ , ENaC, Cl ⁻ , Ca ²⁺ , aquaporins); c. Coupled transporters (Na ⁺ -glucose, Na ⁺ /H ⁺ -antiporter, Na ⁺ -K ⁺ -2Cl ⁻ -symporter, Na ⁺ -phosphate symporter, Na ⁺ -Cl ⁻ -symporter, Na ⁺ -HCO ₃ ⁻ -symporter, Cl ⁻ /HCO ₃ ⁻ -antiporter).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0289	List the nephron sites (PT, LOH, DCT, CT) and explain the molecular mechanisms of action of the following classes of diuretics (carbonic anhydrase inhibitors, loop, thiazide, K ⁺ -sparing).

			SOM.MK.II.BPM1.3.CPR.2.PHYS.0290	For the major solutes (glucose, amino acids, Na, K, Ca, Mg, H ₂ O) of the glomerular filtrate and for PAH and inulin, list the areas of the nephron where these molecules are reabsorbed and secreted.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0291	Explain the effects of diuretics on Na ⁺ handling by the kidneys and, thus, on ECF volume regulation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0292	Outline the effects (up to five) of hormones on the various segments of the nephron.
Lecture 46	PHYS	Distribution of Body Fluids	SOM.MK.II.BPM1.3.CPR.2.PHYS.0293	List major routes and normal ranges for water intake and loss.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0294	Given the body weight estimate the a) total body water, b) extracellular fluid volume, c) intracellular fluid volume, d) interstitial fluid volume, e) plasma volume, and f) blood volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0295	Demonstrate the ability to use the indicator dilution principle to measure plasma volume, blood volume, extracellular fluid volume, and total body water, and identify compounds used to measure each volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0296	Explain the water movement between intracellular and extracellular fluid compartments caused by increases or decreases in extracellular fluid osmolality.
Lecture 47	BCHM	Serum Proteins and Disorders	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318	Classify serum proteins based on electrophoretic mobility. Distinguish the functions of proteins that are found in the alpha ₁ -globulin fraction (alpha ₁ -antitrypsin, alpha ₁ -fetoprotein, retinol binding protein, transcortin).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0319	Predict three common causes of hypoalbuminemia (liver disease, nephrotic syndrome, protein malnutrition) and explain the biochemical basis for the occurrence of edema in hypoalbuminemia.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0320	Explain the pathophysiology of inherited alpha ₁ -antitrypsin deficiency on the liver (gain-of-function) and lungs (loss of function).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0321	Distinguish the functions of proteins in the alpha ₂ -globulin fraction (alpha ₂ macroglobulin, haptoglobin, ceruloplasmin). Describe serum protein changes in nephrotic syndrome and Wilson disease.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0322	Distinguish the functions of proteins found in the beta-globulin fraction (transferrin, hemopexin, beta lipoprotein).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0323	Enumerate the functions of proteins found in the gamma-globulin fraction (Immunoglobulins: IgG, IgM, IgA, IgD, IgE).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0324	Identify serum proteins that are increased following an inflammation. Give examples of positive and negative acute phase proteins.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0325	Interpret the utility of C-reactive protein as an inflammatory marker. Interpret the utility of haptoglobin levels as indicators of hemolysis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0326	Recognize serum protein electrophoretic pattern in the following disorders: a. Acute and chronic inflammation (polyclonal gammopathy) b. Liver cirrhosis c. Monoclonal gammopathy (multiple myeloma) d. Nephrotic syndrome e. Hypogammaglobulinemia f. Alpha ₁ - antitrypsin deficiency
DLA 19	PHYS	Solute Handling	SOM.MK.II.BPM1.3.CPR.2.PHYS.0297	Define the composition of Na ⁺ in extracellular and intracellular fluid volume and its role in maintaining extracellular fluid.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0298	Calculate the normal filtered load of Na ⁺ . List the tubular sites of Na ⁺ reabsorption, and the alterations in Na ⁺ reabsorption in conditions of euolemia, volume depletion, and volume expansion.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0299	Describe the regulation of Na ⁺ reabsorption along the nephron, including the effects of sympathetic nerves, angiotensin II, aldosterone, and atrial natriuretic peptide.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0300	Explain the generation of angiotensin II, beginning with renin. Identify factors that promote renin release.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0301	Describe the role of the renin-angiotensin-aldosterone system in the regulation of systemic arterial blood pressure in volume-replete and volume-depleted states and in secondary forms of hypertension.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0302	Describe the role of the renin-angiotensin-aldosterone systems in the regulation of sodium balance and arterial pressure with emphasis on the actions of angiotensin II on renal hemodynamics and tubular transport.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0303	Describe K ⁺ distribution within the body, extrarenal K ⁺ homeostasis, and the role insulin, epinephrine, and aldosterone play in the movement of K ⁺ between intracellular and extracellular pools. Describe the K ⁺ shift caused by acidosis.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0304	Identify the tubular sites of K ⁺ reabsorption and secretion.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0305	Calculate filtered load of K ⁺ . <i>Describe the regulation of K⁺ secretion by aldosterone in the collecting duct.</i>
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0376	Explain the influence of tubular flow rate, acid-base disturbance and anion delivery on K secretion.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0306	Identify the tubular sites of action of K ⁺ sparing diuretics.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0307	Identify the normal range of dietary calcium and phosphate intake. Describe the renal regulation of plasma calcium by calcitonin and phosphate by parathyroid hormone.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0308	Identify the tubular sites of Ca ²⁺ and phosphate reabsorption.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0310	Describe the role of the kidney in the production of 1,25-dihydroxy vitamin D (calcitriol).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0311	Describe the effects of diuretics on Ca ²⁺ and phosphate excretion, especially noting the effect of thiazides to decrease Ca ²⁺ excretion and loop diuretics to increase Ca ²⁺ excretion.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0312	Identify the tubular sites of Mg ²⁺ reabsorption and describe the regulation of Mg ²⁺ by loop diuretics.

Lecture 48	PHYS	Concentration & Dilution of Urine	SOM.MK.II.BPM1.3.CPR.2.PHYS.0323	Predict the changes in body fluid volume and osmolality caused by a net NaCl or water loss/gain in the body.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0377	Predict how changes in body fluid volume and osmolality alter the rate of urine production and the osmotic composition of the urine.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0324	Identify the role of ADH release that allows either a dilute or concentrated urine to be produced and excreted.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0325	Describe the syndrome of inappropriate ADH release (SIADH) and contrast its mechanism of ADH release to normal stimuli.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0326	Explain the role of thick ascending limb of the loop of Henle in producing a high renal interstitial fluid osmolality.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0378	Beginning with the loop of Henle, compare & contrast the tubular fluid and interstitial fluid osmolality changes that allow either a dilute or concentrated urine to be produced and excreted.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0327	Explain the tubular sites and cellular mechanism by which ADH increases permeability to water and urea.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0379	Describe the role of ADH in making the kidney to produce either a dilute or concentrated urine.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0328	Distinguish between central and nephrogenic diabetes insipidus based on plasma ADH levels and the response to an injection of ADH.
DLA 20	PHYS	Tubular Fluid and Plasma Concentration Profiles	SOM.MK.II.BPM1.3.CPR.2.PHYS.0329	Schematically illustrate and predict the TF/P profiles for Na ⁺ , K ⁺ , Cl ⁻ , HCO ₃ ⁻ , inulin, creatinine, urea, PAH, amino acids and glucose.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0380	Predict how the presence or absence of ADH affects each TF/P profile.
Lecture 49	PHYS	Renal Clearance	SOM.MK.II.BPM1.3.CPR.2.PHYS.0330	Explain the clearance principle. Use the clearance equation and identify appropriate compounds used to estimate the glomerular filtration rate and renal plasma flow.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0331	Distinguish between the use of inulin and creatinine clearances as measures of the glomerular filtration rate.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0332	Given the plasma and urine concentrations and the urine flow rate, calculate the filtered load, tubular transport, excretion rate, and clearance of inulin, creatinine, para-amino hippuric acid (PAH), glucose. Predict how changes in filtration, reabsorption, and secretion will affect renal excretion of each compound.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0333	Discuss the renal handling of glucose under normal conditions and in patients with diabetes mellitus.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0334	Interpret clearance graphs of para-amino hippuric acid (PAH) and glucose at various plasma concentrations of these substances.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0381	With the help of a graph, explain tubular load, tubular transport maximum (T _{max}), renal threshold and slope for glucose.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0335	Discuss the renal handling of PAH under normal conditions and its role in renal in renal plasma flow.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0336	Calculate osmolar and free water clearance & explain its significance. Explain expected free water clearance for an individual producing either a dilute or concentrated urine.
Lecture 50	BCHM	Acid Base Balance: Buffers and Respiratory System	SOM.MK.I.BPM1.3.CPR.1.INTG. 0001	Define an acid and a base and distinguish between strong and weak acids and bases. Define pH, pK _a and buffers.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0003	Analyze the titration curve of a buffer (identify the pK and regions of maximum buffering capacity of a buffer) using the Henderson-Hasselbalch equation.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0004	Distinguish the sources, amounts formed and routes of loss of: volatile acid (CO ₂ derived) and nonvolatile acids
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0005	Analyze the role of HCO ₃ ⁻ as a buffer (its formation and role in buffering) using a graph.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0006	Identify the normal range of pH values as well as the upper and lower limits compatible with life. Emphasize the importance of the 20:1 ratio for maintenance of plasma pH.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0007	Describe the role of hemoglobin as a buffer in CO ₂ transport
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0008	Summarize the role of lungs in maintaining the arterial PCO ₂ .
Lecture 51	BCHM	Acid Base Balance: Role of Kidney	SOM.MK.I.BPM1.3.CPR.1.INTG. 0009	Summarize and analyze the role of kidney in maintaining serum bicarbonate levels.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0010	Differentiate the processes of reabsorption of filtered HCO ₃ ⁻ , formation of new HCO ₃ ⁻ and secretion of HCO ₃ ⁻ in renal tubule.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0011	Explain net acid excretion by the kidneys, titratable acid, and the production and excretion of ammonium.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0012	Explain the process of H ⁺ secretion in renal tubules by the carbonic anhydrase system. Identify factors affecting this process.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0013	Discriminate the role of phosphate and ammonium buffers in the renal tubule. Highlight the significance of glutaminase in prolonged acidosis.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0014	Given a sudden increase or decrease in pH, identify the magnitude and the time course of the compensations that act to minimize changes in pH of the body fluids, including: a) buffers, b) respiratory adjustments, and c) renal adjustments.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0015	Describe the interactions between K ⁺ and H ⁺ and how this could cause acid-base disturbances.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0016	Describe the effects of carbonic anhydrase inhibitors (acetazolamide) and the other diuretics on acid-base balance and the reabsorption of HCO ₃ ⁻ by the nephron.

Lecture 52	BCHM	Acid-base Disorders	SOM.MK.I.BPM1.3.CPR.1.INTG. 0017	Describe processes that lead to respiratory acid-base disturbances (acidosis and alkalosis) and identify common causes for each.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0018	Describe processes that lead to metabolic acid-base disturbances (acidosis and alkalosis). Identify common causes for each including (diabetic ketoacidosis, lactic acidosis, diarrhea, acetazolamide treatment and Vomiting, pyloric stenosis).
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0019	Define and calculate the anion gap and distinguish between increased and normal anion gap in metabolic acidosis.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0020	Describe the derivation of the Davenport diagram and identify how it can be used to determine the acid base status in individuals.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0021	Differentiate between metabolic/respiratory acidosis and alkalosis using the Davenport diagram.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0022	Analyze the compensatory mechanisms that are active in the four acid base disorders and how compensation is represented on the Davenport diagram.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0023	Identify simple and mixed metabolic and respiratory acid –base disturbances using the Davenport diagram.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0024	Distinguish between acute and chronic respiratory disturbances. Describe the causes, compensatory responses for: Acute and Chronic Respiratory acidosis – Causes of hypoventilation. Acute and Chronic Respiratory alkalosis – Causes of hyperventilation, high altitude adaptations.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0025	Interpret ABG data (pH, PCO ₂ and HCO ₃ ⁻) and solve problems related to disorders of acid base balance using Davenport diagram. From blood values, identify simple and mixed metabolic and respiratory acid-base disturbances.
Lecture 53	BCHM	Flipped Class: Interpreting Acid Base Disorders	SOM.MK.I.BPM1.3.CPR.1.INTG. 0024	Distinguish between acute and chronic respiratory disturbances. Describe the causes, compensatory responses for: Acute and Chronic Respiratory acidosis – Causes of hypoventilation. Acute and Chronic Respiratory alkalosis – Causes of hyperventilation, high altitude adaptations.
			SOM.MK.I.BPM1.3.CPR.1.INTG. 0025	Interpret ABG data (pH, PCO ₂ and HCO ₃ ⁻) and solve problems related to disorders of acid base balance using Davenport diagram. From blood values, identify simple and mixed metabolic and respiratory acid-base disturbances.